

A top-down view of a grey desk with various items: a stethoscope, a laptop displaying a website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill-shaped graphic is centered over the laptop.

Pharm*Achieve*

TRANSGENDER HEALTH

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

2SLGBTQIA+ Two-spirit, Lesbian, Gay, Transgender, Queer, Intersex, Asexual, and More

AE Adverse Events

BID Twice Daily

CV Cardiovascular

FSH Follicle-Stimulating Hormone

FTM Female to Male

GAC Gender Affirming Care

GAHT Gender Affirming Hormone Therapy

GnRH Gonadotropin-Releasing Hormone

GnRHa Gonadotropin-Releasing Hormone Analogs

HPG axis Hypothalamic-pituitary-gonadal axis

IM Intramuscularly

LH Luteinizing Hormone

MOA Mechanism of Action

MTF Male to Female

PO By Mouth

RCT Randomized Controlled Trial

SAAB Sex Assigned At Birth

SC Subcutaneously

SL Sublingually

STI Sexually Transmitted Infection

TGD Transgender and Gender-Diverse

UTI Urinary Tract Infection

TRANSGENDER AND GENDER-DIVERSE INDIVIDUALS

Gender/Gender Identity

A person's **self-declared identity** usually based on internal sense of being a male, female, neither, both, or somewhere else along the gender spectrum

Sex/Sex Assigned At Birth (SAAB)

Category assigned/designated at birth based mainly on healthcare provider's exam of external genitalia (e.g. male, female, intersex)

Gender Expression

How one shows their gender identity (e.g. hair style/length, clothing, makeup, body language, voice, pronouns, etc.). May or may not correlate with **gender identity** or **SAAB**

Gender Incongruence

- Marked and persistent experience of **incompatibility** between a person's **gender identity** and their **SAAB**

Transgender and Gender-Diverse (TGD) is an umbrella term used to describe persons whose **gender identity** or **gender expression** is not aligned with their **SAAB** and includes:

- **Transgender:** When **gender identity** is different from **SAAB**
- **Gender diverse individuals:** people whose **gender identity** or **expression** doesn't conform to traditional binary norms of male or female.

ADDITIONAL IDENTITIES

Transgender man/transmasculine describes a person with masculine gender identity who was recorded/designated female sex at birth

Transgender woman/transfeminine describes a person with feminine gender identity who was recorded/designated male sex at birth

Non-binary describes a person who does not identify exclusively as a man nor a woman; may be a combination, something outside these two roles, nonexistent, or fluid

2-Spirit/Two-Spirit are terms used by some Indigenous peoples identifying as diverse gender, sexual, and/or spiritual identities

GENDER-AFFIRMING CARE

Gender Dysphoria

- Distress experienced by a person whose **gender identity** is not aligned with their **SAAB**
- Not all TGD individuals experience gender dysphoria

Gender-Affirming Care

- A range of social, psychological, and medical services designed to support and affirm a person's gender identity, especially when it differs from the sex they were assigned at birth



Other Terms To Know:

- **Cis or cisgender** describes a person with a gender identity that aligns with their SAAB
- Non-binary individuals may use **gender-neutral pronouns** (i.e. they/them/their)
- "*Transsexual*" is an outdated term and should not be used

GOALS OF THERAPY

- ✓ Support the overall health and well-being of TGD individuals through both GAC (if requested) and preventative primary care
- ✓ Empower TGD individuals to make informed decisions about their health
- ✓ Prevent or relieve gender dysphoria when present
- ✓ Provide personalized gender-affirming care that aligns with each individual's goals, preferences, and chosen gender expression

GENDER-AFFIRMING CARE MAY INCLUDE

Social Interventions

- Name, pronouns (e.g. he, she, they, ze or a mix of these)
- Voice-training
- Gender-inclusive language (e.g. body parts, medical procedures)
- Gender presentation changes (i.e. clothing, mannerisms, expressions, nonsurgical body modifications such as tucking, padding, packing, binding)

Medical/Surgical Interventions

- Chest surgery (breast augmentation or mastectomy)
- Genital surgery (vaginoplasty or phalloplasty)
- Hysterectomy
- Orchiectomy
- Feminizing or masculinizing facial surgery
- Laryngeal surgery

PHARMACOLOGICAL MANAGEMENT

Puberty Suppression

- TGD youth may undergo an initial puberty suppression phase before undergoing hormone-based therapy
- Benefits and risks should be discussed with provider

Hormone-Based Therapy (Estrogen-Based or Testosterone-Based)

- Lower endogenous sex hormone levels and/or add exogenous sex hormones to match the individual's gender identity
- Non-binary patients may also use hormone-based therapies to better align with their gender identity, though the dosing may be lower
- Estrogen-Based Therapy
 - Reduce the endogenous effects of testosterone, induce feminine secondary sex characteristics
- Testosterone-Based Therapy
 - Develop androgenic sex characteristics

CONSIDERATIONS

FYI

Consideration #1:

- Some gender-affirming therapies are irreversible, difficult to reverse, or have lifelong effects
- According to a large-scale study of transgender people, 8% reverted back at some point to their sex at birth. Among the reasons for reverting back, external pressures were often cited
- At what age do individuals have the capacity to consent to such potentially permanent medical decisions, particularly if their gender identity may evolve?

Consideration #2:

- The use of puberty blockers and subsequent medical transitioning for minors with gender dysphoria is a prominent area of ethical and scientific controversy
- There is a lack of high-quality evidence in this area
- Conversely, it may not be feasible or ethical to conduct RCTs in this field
- Future areas of research include potential impacts on bone density development, fertility and long-term effects on cognitive, emotional, and psychosocial development during adolescence

CONSIDERATIONS

FYI

Consideration #3:

- Diverging Regulatory Approaches (as of May 2025):
 - Canada:
 - Alberta: Banned puberty blockers for those 15 and under for gender transition
 - Other Provinces: No current outright bans; access typically guided by evolving clinical guidelines and consent processes (many of which require parental, physician and/or psychologist approval)
 - International Policy Divergence:
 - ~25 US States (restrictions/bans);
 - UK (NHS England) (research-only);
 - Finland & Sweden (more cautious guidelines)

PHARMACIST'S TYPICAL ROLES

1. Create a Safe and Inclusive Pharmacy Environment:

- Welcoming Space: Create a welcoming environment for transgender patients including using inclusive language, respecting preferred names and pronouns, and displaying signs that signal support for the 2SLGBTQIA+ community
- Confidentiality and Respect: Maintain patient confidentiality and treat every individual with respect
- Do not impose your own beliefs

2. Medication Management and Monitoring (including hormone therapy):

- Advise prescribers on optimal therapy
- Dispense medication
- Identify and prevent potential drug interactions between hormone therapy and other medications
- Monitor patients for the effects and adverse effects of hormone therapy, in collaboration with other healthcare providers; this may involve ordering and reviewing lab results in jurisdictions that allow it

3. Education and Counselling:

- Counsel patients about their medications, including expected effects, potential risks, and how to manage adverse effects
- Address patients' concerns and questions about their treatment in a safe and confidential manner
- Promote adherence through education and ongoing support

PHARMACOLOGICAL MEASURES

Puberty Suppression (Gonadotropin-Releasing Hormone Analogs)

Drug	Comments
Leuprolide <i>e.g. Lupron Depot®, Eligard®</i>	• Off-label use in pediatrics • MOA: Down-regulates response to endogenous GnRH and sustained suppression of LH and FSH, resulting in the cessation of sex hormone (i.e. testosterone/estrogen) production to prevent further puberty development • AE: Injection site reactions, weight gain, depression/anxiety/mood changes, headache, visual changes, hot flashes • Reversible suppression (puberty typically resumes within 6 months of discontinuation) • May impact fertility potential, bone mineral density • Typically given as IM or SC injection at 1, 3, or 6 month intervals • Dosing intervals may vary depending on patient, prescriber experience, clinical response

PHARMACOLOGICAL MEASURES

Estrogen-based Therapy (Antiandrogens)

Drug	Comments
Spironolactone	• MOA: Potassium-sparing diuretic that also has direct antiandrogen activity (in addition to weak estrogen and progestogen activity) • AE: Hyperkalemia, increased urination, decreased blood pressure, orthostatic hypotension, increased skin sensitivity to sunlight • After orchiectomy, can be stopped or tapered over 4 to 6 weeks • Typically dosed 50 mg PO daily or BID up to a daily maximum of 200-300 mg
Cyproterone acetate	• MOA: Creates negative feedback on the HPG axis to decrease testicular testosterone secretion, and competitively blocks androgen receptors • AE: fatigue, lethargy, dose-related risks for acute liver damage, meningiomas • May be associated with hyperprolactinemia and prolactinoma; monitor prolactin if hyperprolactinemia symptoms are present • After orchiectomy, can be stopped immediately • Taken daily or every other day. Typically dosed 12.5 – 25 mg daily up to a daily maximum of 50 mg PO

PHARMACOLOGICAL MEASURES

Estrogen-based Therapy (Estrogens)

Drug	Comments
Oral/ sublingual 17- beta estradiol	• AE: Headache, migraine, hot flashes, venous thromboembolism, weight gain, mood changes • 17-beta estradiol has reduced thrombogenicity compared to synthetic/conjugated estrogens (consider monitoring estradiol levels) • Typically dosed 1-2 mg PO/SL daily initially then increased up to a maximum of 3 mg PO/SL BID
Transdermal 17- beta estradiol <i>e.g. Estradot®, Climara®, Estrogel®</i>	• AE: Application site irritation, headache, migraine, hot flashes, venous thromboembolism, weight gain, mood changes • Transdermal formulations avoid first-pass metabolism and may be less thrombogenic than oral therapy; considered to have a better safety profile than oral/sublingual routes • Patch: Typically dosed 25–50 mcg/day then increased up to a maximum of 200 mcg/day • Gel: Typically dosed 2.5 g (2 pumps) daily then increased up to 6.25 g (5 pumps) daily
Injectable estradiol valerate	• AE: Injection site irritation/pain, headache, migraine, hot flashes, venous thromboembolism, weight gain, mood changes • No commercially available product; can be compounded • Typically dosed 3-4 mg IM/SC weekly then increased up to 10 mg weekly

GENDER AFFIRMING EFFECTS OF ESTROGEN

Effect	Reversible	Onset	Max effect
Decreased oiliness of skin	Yes	3 months	Variable
Fat redistribution	Yes	3 months	2-3 years
Decreased muscle mass	Yes	3 months	1-3 years
Thinner body hair	Yes	6 months	>3 years
Slowed hair loss	Yes	1 month	Variable
Breast growth	No	3 months	1-2 years
Decreased testicle mass	Possibly	3 months	2-3 years
Decreased libido	Possibly	1 month	3-6 months
Decreased erections	Possibly	1 month	3-6 months
Decreased sperm	Possibly	Variable	Variable
Decreased function of erection	Possibly	Variable	Variable

PHARMACOLOGICAL MEASURES

Testosterone Therapy (Testosterones)



Testosterone therapy is **NOT** sufficient contraception. Patients must be counseled to use additional contraceptives, if needed

Drug	Comments
Injectable testosterone cypionate Injectable testosterone enanthate	• Testosterone cypionate available as 100 mg/mL; compounded in cottonseed oil • Testosterone enanthate available as 200 mg/mL; compounded in sesame oil • AE: Acne, alopecia, insomnia, N/V/D, hypertension, change in libido, injection site inflammation, bladder irritability, weight gain, depression/anxiety • Typically dosed 20–50 mg IM/SC weekly initially then increased up to a maximum of 100 mg IM/SC weekly
Transdermal testosterone gel <i>e.g. Androgel® , Testim®</i>	• Available as a 1% gel • AE: Application site irritation, acne, alopecia, headache, hypertension, depression • Applied once daily to clean, dry, intact skin (shoulders, upper arms, or abdomen) • Should be used at the same time each day, avoid showering (approximately 5 to 6 hours) after application, avoid intimacy until the gel dries (may expose partner to testosterone), and wash hands thoroughly after use

GENDER AFFIRMING EFFECTS OF TESTOSTERONE

Effect	Reversible	Onset	Max effect
Increased skin oil/acne	Yes	1 month	1-2 years
Fat redistribution	Yes	1 month	2-5 years
Increased muscle mass	Yes	6 months	2-5 years
Increase in facial and body hair	No	3 months	4-5 years
Scalp hair loss	No	6 months	Variable
Amenorrhea	Yes	1 month	n/a
Clitoral growth	No	3 months	1-2 years
Vaginal atrophy	Yes	1 month	1-2 years
Voice deepening	No	6 months	1-2 years
Infertility	Variable	Variable	Variable

PREVENTATIVE CARE FOR TRANSGENDER PATIENTS

- Multiple healthcare guidelines and professionals agree that TGD patients are an underserved population when it comes to primary care
 - This could be a result of lack of education and knowledge, the belief that TGD healthcare should be managed solely by specialists, or TGD individuals having negative healthcare experiences in the past
- Pharmacists are uniquely positioned to offer consultation on age-appropriate screenings and monitoring, vaccinations, minor ailments, and other primary care topics to these patients
 - Recommended cancer screenings
 - Recommended vaccinations across the lifespan (shingles, pneumonia, flu, COVID-19, etc.)
 - Minor ailments (UTIs, acne, etc.)
 - Recommended laboratory tests (for monitoring drug therapy and general health)

TRANSFEMININE CANCER SCREENINGS

Cancer Type	Notes
Prostate	The prostate is generally not removed during orchiectomy. Patients with a prostate should receive cancer screenings beginning at age 50.
Testicular	If a patient does not undergo orchiectomy, an annual screening for testicular cancer should be done
Breast	Transwomen over the age of 50 who have been on gender-affirming therapy for 5-10 years should receive a mammogram every 2 years
Neovaginal	Transwomen who have undergone vaginoplasty should receive annual digital and speculum exams of the neovagina
Colorectal	All individuals 50 or older should receive routine colorectal cancer screenings (or earlier depending on family and personal history)



May vary by province or territory – focus on knowing which kinds of cancer screenings are recommended versus the exact ages

TRANSMASCULINE CANCER SCREENINGS

Cancer Type	Notes
Breast	Patients that have not undergone bilateral mastectomy should follow the guidelines for cisgender women. Patients that have had a bilateral mastectomy should have their chest examined routinely.
Endometrial/ Uterine	Not routinely recommended in cisgender women or transmen – transmen that have not undergone hysterectomy should report any abnormal bleeding to their provider
Cervical	Annual pap smears are recommended in transmen that have a cervix and are over the age of 21
Ovarian	Not routinely recommended in cisgender women or transmen – transmen that have not had an oophorectomy should report any concerning symptoms to their provider
Colorectal	All individuals 50 or older should receive routine colorectal cancer screenings (or earlier depending on family and personal history)



May vary by province or territory – focus on knowing which kinds of cancer screenings are recommended versus the exact ages

PRIMARY CARE CONSIDERATIONS FOR TGD PATIENTS

Condition	Management
UTI	Increased frequency in patients that have had genital surgery Genital surgery is considered a major alteration to the GU anatomy, making UTIs in these patients considered to be complicated cystitis
STIs	Anatomy of patient will determine which testing and treatments should be completed Patients that have undergone genital surgery may have different organisms that cause infection compared to patients that have not undergone surgery
Mental Health Concerns	TGD patients may have mental health concerns associated with their gender dysphoria and/or concerns resulting from public discrimination or stigma Important to screen for and treat any patient concerns that arise
Testosterone-associated acne	Testosterone increases sebum production which may be exacerbated by changes in clothing, binders, etc. Patients do not need to stop their GAHT - Follow general cystic acne guidelines - Remember: some therapies require adequate contraception – testosterone is NOT an adequate contraception for this purpose
Cardiovascular Disease	Patients receiving hormone therapy have an increased risk of CVD - Estrogen: Increase risk of stroke and other clots, changes to blood pressure and lipids, etc. - Testosterone: Increased hemoglobin, changes to blood pressure and lipids, etc.

MONITORING

Efficacy Endpoints

Parameter	Degree of Change	Timeframe
Gender dysphoria	Decrease	Ongoing
Masculinizing or feminizing effects	Increase	
Mental health and wellbeing	Improve	
Testosterone or estrogen levels	Maintain in efficacious but safe range	

MONITORING

Safety Endpoints		
Parameter	Degree of change	Timeframe
Medication AEs	Prevent or minimize	Throughout duration of therapy
Blood pressure, lipids, CV risk factors	Maintain normal levels	
Surgical complications (UTIs, bleeding, etc.)	Prevent	During and post surgery

TAKEAWAY

- Gender identity may not align with an individual's SAAB
- Transgender and Gender-Diverse (TGD): persons whose gender identity is not aligned with their SAAB
- Puberty suppression involves the use of GnRHa's and is reversible, with puberty/hormonal changes typically resuming within 6 months of discontinuation
- Hormone-based therapy involves lowering endogenous sex hormone levels and/or adding exogenous sex hormones to match the individual's gender identity
 - Estrogen-based (using antiandrogens and/or estrogen)
 - Testosterone-based (using testosterone)
- Some effects of hormone therapy are irreversible – informed consent should be given by the patient prior to initiation of therapy
- Pharmacists are in an excellent position to assist in the primary care of TGD patients including recommending cancer screenings, vaccinations, minor ailments treatments, and monitoring of both general health and GAHT

CASE 1

DS is a 26-year-old transgender woman who is beginning estrogen-based hormone therapy. DS experiences marked and persistent gender incongruence and has adopted several social changes as part of her transition, including a name change, adopting "she/her" pronouns, wearing more feminine clothing and hairstyles, and tucking. DS has a history of depression and anxiety, which is managed by sertraline 100 mg PO once daily and hypertension managed by candesartan 16 mg PO once daily. DS has not undergone any medical/surgical changes and is excited but anxious about this next step in her gender-affirming care. DS currently drinks 2 cups of coffee a day and smokes ½ a pack of cigarettes a day. Her body mass index is 19 kg/m². DS' labs are unremarkable, though her potassium is on the upper limit of normal according to her physician.

DS presents a prescription for 17-beta estradiol 1 mg PO once daily x 4 weeks.

1. DS has changed her first name to "Devon", but her government health card still states "Daniel". When speaking to DS, which name should be used?
 - a) Daniel, because this is her legal name
 - b) Devon, because this is her name
 - c) Avoid using first names, because this is confusing
 - d) Ask DS what name to use each time, because she may change her mind

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2. Which of the following adverse effects is the LEAST likely to result from estrogen therapy?
- a) Hypotension
 - b) Hot flashes
 - c) Weight gain
 - d) Mood changes

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3. All of the following are appropriate parameters to monitor for DS, EXCEPT:
- a) Serum estradiol
 - b) Worsening anxiety, mood changes
 - c) Appropriate signs of feminization
 - d) Serum sodium

CASE 1

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DS presents a prescription for 17-beta estradiol 1 mg PO once daily x 4 weeks.

4. DS wants to know if there are any other medications that might be prescribed for gender affirming care. Which of the following would be the BEST medication to suggest for DS?
- a) Progesterone
 - b) Spironolactone
 - c) Cyproterone acetate
 - d) Estradiol valerate injection

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CHANGE LOG

August 2025

- New lecture

November 2025

- Updated legend
- Slides 22 and 23: minor edits

March 2026

- Slide 2: Updated NAPRA competencies
- Slide 3: Updated legend
- Slide 14: Minor changes
- Slide 17: Removed mention of depo-testosterone® as no longer available on Canadian market